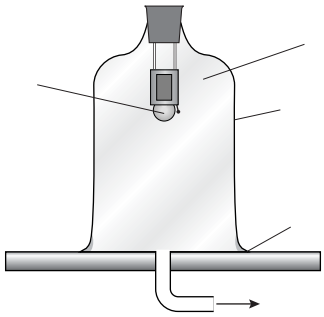
# How does sound travel through solids, liquids and gases?

This diagram shows an experiment to show that sound cannot travel through empty space.



**1** Complete the diagram by adding these labels:

| | | |
|---|---|---|
| **vacuum** | **bell** | **to pump** |
| | **glass jar** | **grease** |

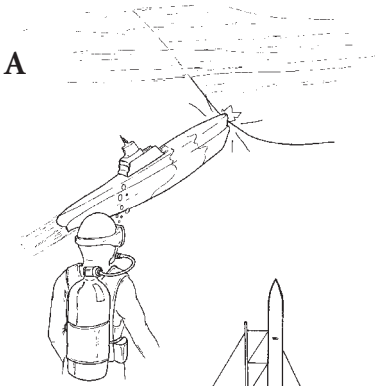
**2** Complete this description by filling in the gaps.

Before the pump is turned on we can hear the bell. The sound travels through the air inside the jar, and through the air in the room to our ears. This tells us that sound travels through a _____. The sound travels through the jar, which is made of _____. This tells us that sound travels through _____.

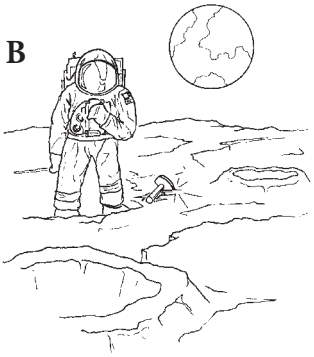
After the pump has been on for a while, we _____ hear the bell. This tells us that sound cannot travel through a _____.

**3** These people can see what is happening, but can they hear it? Tick the ones who can hear, and put a cross by the ones who can't.

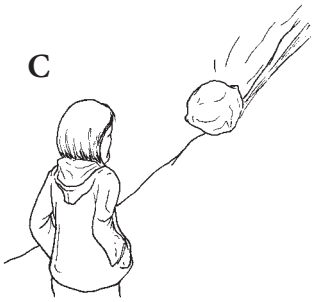
A



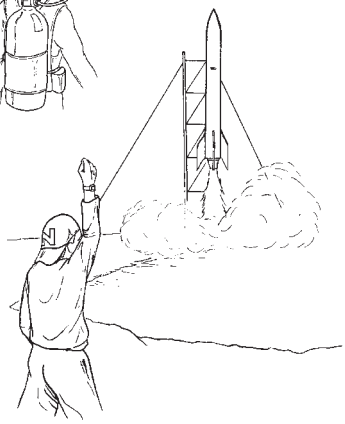
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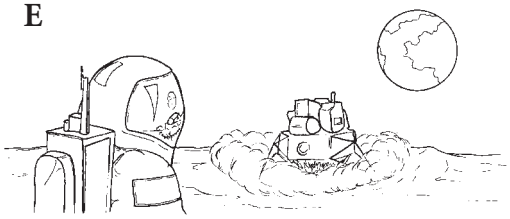
C



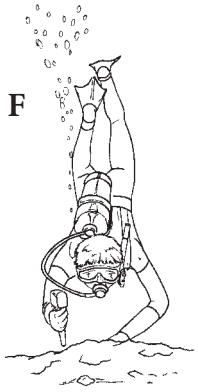
D



E



F

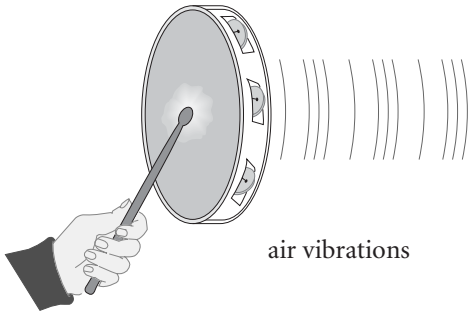




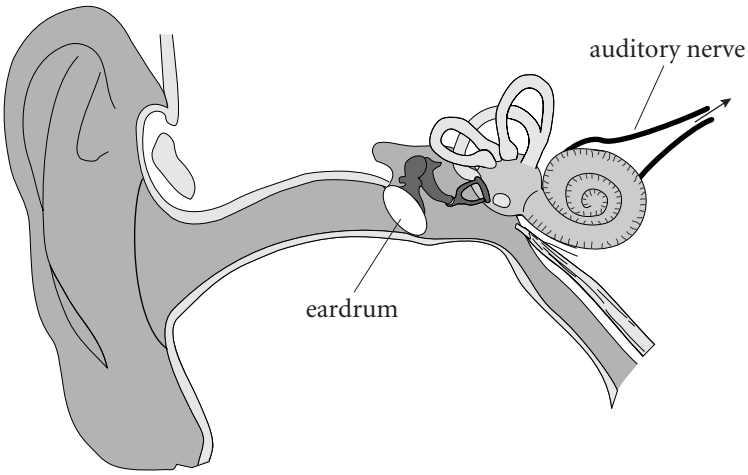
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# How do we hear sounds?

**1** Look at the diagrams and use them to help you complete the sentences below.




**a** When a drum is hit, the surface starts to _____. This starts the _____ vibrating so that sound travels through the room.




**b** When the sound enters the _____, the air vibrates against the eardrum.

The surface of the eardrum starts to _____.

These vibrations are transmitted through bones and liquid to the nerves.

The _____ _____ sends messages to the brain.

**2** When sound is reflected from something, we call it an echo.
Bats make very high-pitched squeaks when they fly, and they listen for an echo.
Bats can fly in the dark without banging into things using echo location.

**a** If a bat heard an echo of the squeak, what would this mean?

_____

**b** If a bat did not hear an echo, what would this mean?

_____



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# Can sound be dangerous?

When you hear a **loud** sound, the vibrations have a **large amplitude**.

The vibrations transmitted through the ear will be so large that they may damage the little hairs inside that detect the sound.

The longer the sound continues, the more likely the hairs are to be damaged, and the less likely they are to recover afterwards.

This would mean that you wouldn't be able to hear as well.

If you couldn't hear the things in the table below, you would have problems.

**1** Complete this table describing the problems. (You may want to use the word 'isolated' – it means set apart or separated from everyone else.)

| **Sound** | **Problem caused by not hearing the sound** |
|---|---|
| car horn | |
| bicycle bell | |
| smoke alarm | |
| telephone ringing | |
| doorbell | |
| voices on the phone | |
| people talking | |
| television | |

**2** Here are some actions people could take. Put a tick by the ones that would help to protect their hearing.

**a** Jade is at a rock concert. She could protect her hearing by putting cotton wool in them, or she could:

☐ 1 sing along to the music ☐ 2 have an alcoholic drink

☐ 3 move away from the speakers ☐ 4 take some paracetamol.

**b** Jack drives a tractor. He could protect his hearing by:

☐ 1 wearing ear defenders ☐ 2 listening to the radio

☐ 3 taking aspirin ☐ 4 chewing gum.



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# Investigating sound

## Word search

| a | m | p | l | i | t | u | d | e | f | x | u | s | t | y |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| b | j | r | f | l | n | o | i | s | e | h | l | o | w | b |
| m | r | t | n | t | j | l | o | u | c | k | p | u | f | b |
| t | j | i | f | l | s | f | r | e | q | u | e | n | c | y |
| x | l | o | u | d | w | a | m | r | s | z | i | d | p | v |
| c | s | v | t | r | e | y | o | j | a | b | r | t | i | k |
| i | o | v | p | y | t | n | w | h | i | g | h | b | h | o |
| n | j | o | i | r | b | l | d | e | a | f | b | l | e | t |
| o | x | f | t | b | h | y | a | r | l | n | v | c | a | d |
| s | u | v | c | k | r | b | o | t | o | l | n | j | r | s |
| a | e | c | h | o | n | a | i | z | u | w | a | o | p | h |
| r | i | o | y | w | i | m | t | o | b | n | d | r | u | m |
| t | o | c | q | u | m | u | s | i | c | g | z | p | o | l |
| l | o | h | y | d | r | o | p | h | o | n | e | g | n | y |
| u | x | l | w | d | e | m | g | e | s | n | x | e | p | c |
| o | d | e | c | i | b | e | l | n | u | m | f | o | r | d |
| b | e | a | v | j | o | t | i | n | n | i | t | u | s | t |

Look for all the words about sound in the lists below. Circle them on the grid.

**amplitude**
**sound**
**noise**
**low**
**soft**
**loud**
**frequency**
**ear**
**ultrasonic**
**vibration**
**high**
**deaf**
**hear**
**pitch**
**hertz**
**echo**
**music**
**drum**
**cochlea**
**hydrophone**
**decibel**
**tinnitus**



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